

GRAFT: Symmetry-Aware Atom Mapping through Continuous Fragment Growth

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Atom-to-atom mapping for reaction analysis requires retaining alternative correspondences without enumerating redundant atom permutations. We introduce GRAFT, a graph-based method that grows matched fragments on weighted reactant and product graphs, represents compatible placements through correlated symmetry actions, and branches on distinct saturated fragment placements. The default pipeline uses one seed ordering per cut and the cut sweep. Completed searches are deduplicated by unordered matched fragment pairs while preserving their internal mapping alternatives. A symbolic decoder extracts distinct signed bond-event patterns directly from these families, separating the tolerance used for matching from the thresholds used to interpret bond changes. On all 1,851 Golden benchmark records, bidirectional sweep search recovers an annotated correspondence among its retained alternatives in 1,834 cases (99.08%) with one seed ordering per cut (default) and 1,840 (99.41%) with ten. All GRAFT reference-recovery verdicts are resolved. The strongest evaluated prior-method alternative-coverage baseline, the bidirectional SLAP sweep, recovers 97.03%. On the matched 1,821-reaction timing cohort, the one-seed GRAFT search stage and SLAP sweep average 2.52 and 4.82 CPU seconds per reaction, respectively. On 140 coordinate/Wiberg-bond-order reaction pairs, GRAFT with a 2,000-branch cap reaches the same minimum event count as the SLAP sweep in 136 reactions and fewer events in four. Where their minima agree, it retains 158 of 164 comparator minimum-event patterns. At the default cap of 100, 138 reactions return complete mappings and two return none. This bond-event comparison uses unverified mappings and does not measure mapping accuracy. GRAFT provides an inspectable representation of mapping and bond-event alternatives for subsequent chemical assessment.

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1 Introduction

Atom-to-atom mapping assigns corresponding atoms across a reaction. It underlies reaction-center identification, template extraction, and the preparation of endpoint structures for mechanistic calculations. A reaction can admit several plausible correspondences, while many different atom-index assignments are chemically equivalent. An algorithm intended to retain alternatives must therefore resolve two questions: which structural choices deserve separate candidates, and which permutations can be represented together?

Existing approaches use learned correspondence signals or graph-based optimization. RXNMapper derives mappings from attention in a reaction language model (11); LocalMapper learns from chemist-labeled reactions (2); and SLAPMapper combines iterative label refinement with sequential linear assignment (7). The Golden benchmark provides a common set of annotated reactions for assessing mapping recovery (8). For applications requiring alternatives, recovering the annotation somewhere in a returned family is a different objective from producing one correct top-ranked mapping.

Weighted endpoint graphs introduce a further distinction. A permissive matching tolerance can preserve a fragment despite changes in its bond orders, but the corresponding atom shuffles need not preserve the bond events assigned at a finer threshold. Replacing a compressed family by one representative can therefore hide distinct transformations. Likewise, greedily growing one fragment can prevent a neighboring fragment from reaching an alternative assignment.

GRAFT addresses these issues by branching over structural fragment choices while keeping symmetry-related atom bijections compressed (Figure 1). Continuous growth and alternative placements determine the structural choices. Default GRAFT uses one seed ordering per cut and the cut sweep. We introduce a cut-sweep strategy that changes the search conditions to counteract greedy fragment commitment; it is part of default GRAFT, and we also apply it to SLAP to test its benefit beyond our fragment search. Its final representation groups unordered matched fragment pairs and retains the allowed mapping alternatives within each group. A constraint-based decoder projects these families onto distinct bond-event patterns. The method searches for structurally compatible alternatives without requiring the reference transformation to minimize the number of bond changes.

2 Method

Algorithm 1 connects the fragment-level operations in Figure 1. A live branch stores the fragments already placed, their atom assignments, and the compressed choices still allowed by those decisions. Growing and placing the next fragment conditions every later continuation. The matching routine is treated as an operation here; its compatibility tests and symmetry representation are defined below. For bidirectional search, the search phase is applied in each orientation, retaining the orientation of each returned family.

The forks in the search phase represent different fragment placements, not one branch for each symmetry-related bijection. The cut sweep changes the conditions of this search; a single mask can expose a mapping with several bond events. Event decoding is a separate operation on the retained output. Search caps are recorded as truncation, not treated as proof that no alternatives exist.

2.1 Weighted matching and continuous growth

Let V_R, V_P be the reactant and product atom sets, Z_R, Z_P their element labels, and W_R, W_P their symmetric bond-weight matrices. A partial mapping $m : D \subseteq V_R \rightarrow V_P$ is injective and element compatible. A mapping assigning every source atom is bijective when the endpoint element counts agree. Explicit hydrogen atoms participate in the search. Molecular coordinates can accompany the graphs but do not determine the growth decisions.

The active source graph contains edges with weight at least b . Starting from a seed atom, GRAFT grows a connected source fragment F while carrying multiple compatible product placements. A frontier queue prioritizes stronger source edges. To add an unassigned source atom u with product image v , a placement

Algorithm 1 GRAFT: from fragment growth to alternative bond-event candidates

Require: Oriented endpoint graphs, seed orders, optional anchors, search budgets

Ensure: Compressed final branches; optionally, distinct bond-event candidates

```
1:  $\mathcal{S} \leftarrow \emptyset$  {collected terminal branches}
2: for all conditions: uncut source graph and each single-edge source mask do
3:   for all requested seed orders under this condition do
4:     Initialize live branches with the supplied anchors
5:     repeat
6:       for all seeds in the current order do
7:         for all live branches eligible to grow from this seed do
8:           Grow a fragment to saturation, conditioned on this branch {Fig. 1a}
9:           if no admissible fragment placement is found then
10:            Retain this branch for later seeds
11:           else
12:            Fork one continuation per distinct structural fragment placement
13:            Commit each placement; retain its compressed symmetry and constraints
14:            Admit the continuations within the branch budget
15:           end if
16:         end for
17:       end for
18:     until all branches are finished or a full pass makes no progress
19:     Add terminal branches and their search-stop status to  $\mathcal{S}$ 
20:   end for
21: end for
22: Group final branches by unordered fragment pairs; retain their family union
23: Expose compressed branches with constraints and provenance
24: if bond-event candidates are requested as separate post-processing then
25:   Decode distinct event classes from complete families using the original matrices
26:   Report witnesses, supporting families, and completeness status {Fig. 1c}
27: end if
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must satisfy

$$Z_R(u) = Z_P(v), \quad v \notin m(D), \quad W_P(v, m(r)) \geq b, \quad |W_R(u, r) - W_P(v, m(r))| \leq \tau_{\text{iso}} \quad (1)$$

for every active, nonrelaxed source edge (u, r) into F . Injectivity is enforced against all current assignments in D , including established fragments. All relevant fragment bonds are checked, rather than only the frontier edge that proposed the extension. Source nonedges do not impose negative matching constraints, so additional product bonds remain possible.

An accepted extension enlarges the fragment and filters its compatible placements. Growth continues even if only one placement remains. An unsupported edge is deferred, and other frontier edges can still be explored. When growth reaches an established fragment, ordinary continuation respects its existing assignments. A fragment saturates when no further admissible extension remains; structurally distinct surviving placements then generate separate continuations for the remaining atoms.

This is a greedy search along each trajectory: an accepted extension can remove placements that would support a different fragment boundary. GRAFT broadens coverage through cut sweeps and seed orderings.

2.2 Cut sweep to vary greedy fragment boundaries

Our cut-sweep strategy reruns growth under alternative source-edge conditions. In addition to the uncut graph $G_R = (V_R, E_R)$, it searches a graph with one eligible edge e removed, $G_R^{(e)} = (V_R, E_R \setminus \{e\})$, for each edge at or above the cut threshold. The evaluated cut threshold equals the active-graph floor $b = 0.2$, and eligible edges include bonds to H. Each subrun retains the atom set and the original endpoint weight matrices.

Removing e changes both growth and matching: the frontier cannot extend along that edge, and its bond-preservation test is omitted from Equation (1). An extension accepted in the uncut run may otherwise merge

atoms into a fragment and discard alternative placements. Under the cut condition, that growth route is unavailable; a fragment can stop at a different boundary, grow through another route, or leave atoms for a later seed. The missing preservation test also admits placements previously excluded by e . Thus the sweep counteracts greedy commitment by changing the conditions under which growth proceeds. It does not force the endpoints of e into different final fragments or require a bond-breaking event: they may reconnect through other source edges, and the final correspondence may preserve their bond.

GRAFT takes the union of the uncut and single-edge subruns before final branch deduplication. The default one-seed mode uses one seed ordering per subrun, so it already includes the full cut sweep. There are $|E_R| + 1$ subruns at the evaluated thresholds. A single-edge cut describes the search perturbation, not the number of bond events in a returned mapping. One removed constraint can redirect subsequent fragment growth and placements throughout the reaction, exposing an alternative with several bond formations, cleavages, or bond-order changes. GRAFT therefore need not enumerate combinations of final bond events to discover multi-event alternatives. Bond-event decoding uses the original matrices, so a search cut itself contributes no event or event-count penalty. Growth remains greedy within each conditioned subrun; the union broadens coverage without making the search exhaustive.

2.3 Automorphisms and correlated mapping families

The distinction between structural branches and explicit bijections is central to the design. Even one fragment family can contain combinatorially many assignments: k independently exchangeable methyl-hydrogen triplets admit $(3!)^k$ bijections. Expanding those automorphisms into separate mappings would multiply the work without adding fragment choices. GRAFT instead stores the permitted permutation actions and their constraints, and decodes event alternatives directly from this representation. Branch caps bound retained alternatives within each growth call; they do not guarantee a small global search. This avoids explicitly listing symmetry-related bijections within each retained family.

Compatible placements are stored using a representative assignment, unresolved assignment blocks, and permutation generators. During growth, graph canonicalization accounts for the current fragment, locked assignments, and boundary information. Each saved transition records its action generators together with conditioning and fragment-preservation constraints; their conjunction defines the allowed mapping relation. Canonical labeling and automorphism computation use nauty (9).

Matching tolerance and automorphism equivalence have different roles. The relation $|x - y| \leq \tau_{\text{iso}}$ is generally not transitive, so it does not by itself define a permutation group. To describe an intrinsic symmetry shared by a final fragment pair, we instead color product bonds by their responses to the fragment’s matching tests. For a product weight w and the set $\mathcal{W}$ of active reactant weights for the same element pair, the label records

$$\chi_{\mathcal{W}}(w) = (\mathbf{1}[w \geq b], (\mathbf{1}[w \geq b \wedge |w - v| \leq \tau_{\text{iso}}])_{v \in \mathcal{W}}). \quad (2)$$

An exact automorphism of this colored graph preserves every encoded matching test. This definition depends on the final fragment pair, not its growth order.

The saved mapping relation also contains the constraints imposed by its search decisions. We represent a flat family $\mathcal{F}_j$ by a seed mapping m_j , a sequence of allowed action groups $G_{j,1}, \dots, G_{j,k}$, and final preservation constraints C_j :

$$\mathcal{F}_j = \{a_k \circ \dots \circ a_1 \circ m_j : a_i \in G_{j,i}, C_j(a_k \circ \dots \circ a_1 \circ m_j)\}. \quad (3)$$

These actions encode correlated moves; atoms in a common orbit are not independently permutable. Disjoint actions commute and can be placed in a canonical order. The order of overlapping actions is retained. A sequence of groups is not generally equivalent to the group generated by their union.

The intrinsic automorphism of a fragment pair is shared metadata, not permission to enlarge its saved family. Multiple admissible seed mappings can occupy different intrinsic symmetry orbits, and different search constraints can restrict the allowed actions. GRAFT therefore retains these alternatives explicitly inside the compressed representation.

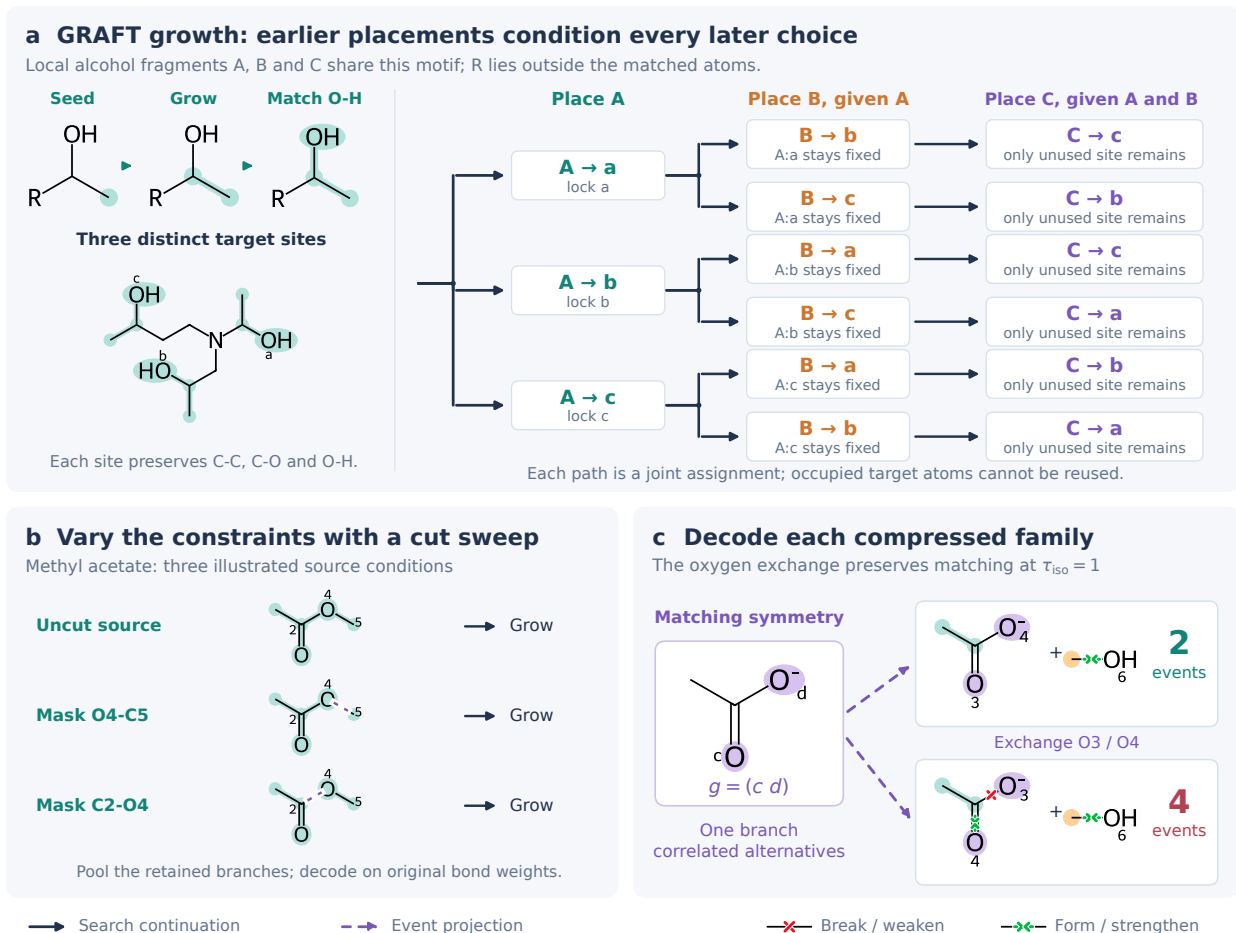


Figure 1 GRAFT: conditional fragment branching and compressed symmetry. (a) A local alcohol motif grows while retaining its O-H bond. Three sites in the depicted target have distinct molecular environments. For three source fragments sharing this motif, placing A fixes one site; B then branches over the two unused sites, and C takes the remaining site. The tree illustrates local conditioning and injectivity; R denotes atoms outside the matched subgraph, and this is not a complete reaction-search trajectory. All attached hydrogens in the illustrated local matches are checked. (b) The cut sweep searches the uncut source graph and graphs with one source edge masked. The illustrated ester masks change the growth and preservation constraints; their outputs are pooled. Masking an edge does not itself assign a bond event. (c) Exchanging the two acetate oxygen images preserves the fragment’s element and bond-weight matching tests at $\tau_{iso} = 1$, but changes two heavy-atom events into four at threshold 0.5. Neither acetate oxygen carries H. Solid black lines retain the endpoint bonds. Red $\times$ marks on the bonds indicate cleavage or weakening; paired green arrows pointing inward indicate formation or strengthening. Pastel colors track fragment membership or symmetry sites. Dashed arrows are event projections within a family, not additional search branches. Saved constraints govern allowed symmetry actions. These formal-bond-order event signatures illustrate GRAFT decoding, not distinct mechanisms or resonance-normalized transformations.

2.4 Unordered final branches

During search, history and boundary conditions matter because they determine valid continuation. At completion, the branch identity is instead the unordered collection of matched atom sets,

$$\mathcal{B} = \{(R_1, P_1), \dots, (R_q, P_q)\}, \quad P_i = m(R_i), \quad (4)$$

under a fixed matching policy. Sorting atoms within each set and then sorting the fragment pairs removes dependence on growth order and fragment labels. Growing A before B or B before A yields one branch when the final paired sets agree.

Each branch stores the union of its flat families. Identical normalized family records are shared, while distinct constraints or action programs remain available. This transformation preserves the represented mapping union: it changes its grouping, not its admissible assignments. Provenance links retain the discovering paths, but the original search hierarchy is unnecessary for decoding. This conservative deduplication does not assert that the remaining families are mutually disjoint.

2.5 Decoding distinct bond-event patterns

For each complete mapping, define the signed change for an unordered source pair $\{u, v\}$ by

$$\Delta_{uv}(m) = W_P(m(u), m(v)) - W_R(u, v), \quad s_{uv}(m) = \begin{cases} +1, & \Delta_{uv}(m) \geq \delta_{uv}, \\ -1, & \Delta_{uv}(m) \leq -\delta_{uv}, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Positive events denote bond formation or strengthening; negative events denote cleavage or weakening. The event count is $E(m) = \sum_{u < v} \mathbf{1}[s_{uv}(m) \neq 0]$. The evaluation includes hydrogen atoms and all atom pairs. For the WBO collection, $\delta_{uv} = 0.3$ for pairs involving a metal and 0.5 otherwise. These thresholds are independent of τ_{iso} .

The decoder returns one mapping witness per distinct signed event class, identifying patterns related by exact source permutations that preserve all thresholded bond-change tests. A permutation preserving fragment matching may therefore produce a distinct event class. First, symmetry-invariance certificates identify families whose mappings all produce equivalent events, and reachable-image bounds exclude families outside a requested event window. Before constraint encoding, adjacent nested action groups are absorbed: if $H \leq G$, then $HG = GH = G$. Exact generator-membership tests remove these redundant choices without changing the final preservation constraints. Remaining families are encoded symbolically in Z3 (5). Each satisfying assignment yields a concrete mapping and its event pattern. A constraint then excludes that entire event-equivalence class, and the process repeats. This projects directly onto event classes rather than explicitly enumerating full mappings or permutation groups.

For a window $E \leq K$, decoding is complete when every retained family is certified exhausted or excluded. A timeout leaves completeness unresolved. This certificate concerns the saved search output within the window; it does not establish exhaustive exploration of all chemically possible mappings. The default deduplication preserves families that contain different event counts, as illustrated in the heavy-atom projection in Figure 1c.

3 Evaluation

3.1 Datasets and configurations

We evaluate annotated mapping recovery on all 1,851 Golden records (8) and event alternatives on 140 native coordinate/WBO endpoint pairs. Golden inputs use formal bond orders and explicit-H expansion. Reference labels are removed before search. Correctness compares the annotated heavy-atom correspondence, including unmatched atoms, modulo endpoint chemical automorphisms preserving charge, isotope, hydrogen count, and stereochemistry. A recovery requires a verified reference witness among the returned alternatives. Unresolved cases remain in the denominator.

We use two explicit prior-method baselines. LocalMapper is a prior accuracy state-of-the-art method as reported by Chen et al. (2); it is also the strongest single-output mapper in our strict Golden re-evaluation (1,605/1,851, 86.71%). Its published Golden accuracy was 89.8% on 1,758 retained reactions with manual calibration, so we keep the literature result separate from our rescored result.

The original SLAP paper reports 86.9% Golden accuracy on the 1,758-reaction version distributed with LocalMapper, counting reference inclusion among returned solutions (7). In our strict 1,851-reaction re-evaluation, default binary SLAP recovers 1,591 references (85.95%). We additionally apply the cut-sweep strategy introduced in this work to SLAP and combine both search directions and binary/weighted modes. This extended configuration recovers 1,796 references (97.03%), making it our strongest evaluated prior-method baseline for alternative recovery and the paired comparator for directional recovery and same-host

Table 1 Evaluation configurations. Both use $b = 0.2$, $\tau_{\text{iso}} = 1$, explicit H, and uncut plus single-edge searches. Default GRAFT uses one seed per cut, the sweep. Golden measures the sweep-search stage; no-sweep controls retain only the uncut graph. A seed setting counts source-atom orderings per cut.

Setting	Golden	Coordinate/WBO
Reactions	1,851	140
Search directions	Both, unioned	Reactant to product
Seed orderings	1 (default), 2, 3, 10	1 (default)
Branch cap	100 (default)	100 (default), 2,000
Evaluation	Reference-family recovery	Signed event-class coverage

search timing. Throughout, “SLAP sweep” denotes SLAP augmented with our strategy, not the original SLAP algorithm or its published score. The sweep’s effect is measured against a paired uncut union with the same directions and modes (Table 2); differences from the literature score cannot be attributed to the sweep alone. Runtime baselines are identified within each recorded hardware and output scope rather than assigning one method a universal speed ranking.

The coordinate collection contains balanced endpoint pairs with 18–137 explicit atoms, including organometallic and multicomponent structures. It has no annotated mappings and was inspected during method development. This is a bond-event comparison, not a mapping-accuracy benchmark: the mappings are unverified. We compare the minimum signed-event count reached by each method and the alternative event patterns at that count. These are minima within the returned candidate sets, not proven global optima or validated mechanisms. SLAP outputs are rescored on the same original WBO matrices using Equation (5). Each returned heavy-atom assignment is completed by symbolic hydrogen refinement. The comparator classes are those observed among these returned witnesses; its hydrogen label families are not exhaustively decoded.

Table 1 separates the settings actually evaluated. Golden measures reference recovery from search; the 140-case evaluation additionally decodes bond-event patterns. Comparator mappings are not used to propose or accept search branches. A two-order run of the final coordinate pipeline was not performed. Full configurations and reproduction scripts are linked in the code repository (Appendix A).

3.2 Golden mapping recovery

With one ordering per cut (default), the bidirectional GRAFT sweep-search stage recovers 1,834/1,851 annotated mappings (99.08%). Three orderings recover 1,837 (99.24%; Table 2), and ten recover 1,840 (99.41%), with all reference-recovery verdicts resolved. Two orderings also recover 1,834 (99.08%), with all reference-recovery verdicts resolved at these settings as well.

A separate same-CPU ablation repeats all 1,851 reactions at caps 100 and 2,000 (Table A3). With the sweep, cap 2,000 gives the same confirmed recovery at one, two, and three orderings, while mean search CPU on the common completed cohort increases by 4.5–5.4 times. Ten orderings at cap 2,000 confirm 1,835 recoveries and leave 12 cases unresolved; five of those were recovered at cap 100, so the lower confirmed count does not establish a loss of representable solutions. No swept setting gains or loses a recovery among reactions resolved at both caps. Without the sweep, the larger cap adds 19 confirmed recoveries, from 1,489 to 1,508. The appendix reports accuracy, unresolved outcomes, and CPU costs for every configuration together with the paired SLAP baselines.

At one seed ordering, smaller-first search recovers 1,803/1,851 references (97.41%), larger-first search recovers 1,779 (96.11%), and bidirectional search (the union of both directions) recovers 1,834 (99.08%; Table 3). Size counts all explicit atoms, including H. On equal-size endpoints, smaller-first retains the original input orientation and larger-first reverses it. Bidirectional search adds 31 recoveries over smaller-first alone. Among the 1,084 strictly unequal-size pairs, the corresponding counts are 1,055, 1,043, and 1,072; among 767 equal-size pairs, they are 748, 736, and 762. Thus the bidirectional gain includes both size regimes.

These are family-recovery measurements: they quantify whether a reference correspondence is retained, rather than whether it is ranked first. Verification processes saved cuts independently, retaining one cut in memory

Table 2 Golden reference recovery and runtime. Recovery uses all 1,851 primary-evaluation records. GRAFT retains compressed families; SLAP returns explicit candidates; other methods return one bijection over their matched atoms. GRAFT uses cap 100 and bidirectional search. Bidirectional SLAP combines both directions and both binary/weighted modes; the default SLAP row uses binary only. Times are mean CPU seconds per reaction, excluding separate bond-event decoding. Unmarked times use the same 1,403 completed reactions on an AMD EPYC 9J14. *a*: archived completed calls (1,718–1,851 per method), CPU model unrecorded; these times are not directly comparable with unmarked times. †: LocalMapper, prior accuracy SOTA (2024). ‡: SLAP with our sweep, strongest evaluated prior-method recovery baseline.

Method	Search setting	Output	Reference recovered	Mean CPU s
GRAFT	1 seed, sweep (default)	Families	1,834 (99.08%)	0.963
GRAFT	2 seeds, sweep	Families	1,834 (99.08%)	1.619
GRAFT	3 seeds, sweep	Families	1,837 (99.24%)	2.265
GRAFT	10 seeds, sweep	Families	1,840 (99.41%)	6.993
GRAFT	1 seed, no sweep	Families	1,489 (80.44%)	0.040
SLAP, bidirectional	No sweep	Candidates	1,661 (89.74%)	0.105
SLAP, bidirectional‡	Sweep	Candidates	1,796 (97.03%)	2.793
SLAP, binary	Default, no sweep	Candidates	1,591 (85.95%)	0.075 ^a
RXNMapper	Default, no sweep	One bijection	1,547 (83.58%)	0.056 ^a
LocalMapper†	Default, no sweep	One bijection	1,605 (86.71%)	0.294 ^a
Chython	Default, no sweep	One bijection	1,561 (84.33%)	0.612 ^a
Indigo	Default, no sweep	One bijection	692 (37.39%)	0.184 ^a
RDT	Default, no sweep	One bijection	1,030 (55.65%)	1.021 ^a

at a time. A reference is absent from the saved union only when every scheduled cut has been checked and excludes it; missing cuts or inconclusive checks remain unresolved.

In the no-sweep ablation, one-seed GRAFT recovers 1,489/1,851 references (80.44%), with no unresolved cases. Uncut bidirectional SLAP, combining binary/weighted modes, recovers 1,661 (89.74%), with one unresolved case. Our cut sweep raises GRAFT recovery from 80.44% to 99.08% (345 additional references) and SLAP recovery from 89.74% to 97.03% (135 additional references). These paired gains demonstrate the strategy’s benefit for both methods under the evaluated protocol. Thus, GRAFT’s higher recovery in the swept comparison depends on the broader search: its one-seed uncut control has lower coverage than uncut bidirectional SLAP. The controls use saved uncut outputs from the same final campaign.

The gain is consistent with GRAFT’s greedy fragment growth. Once an extension is accepted, placements incompatible with that larger fragment are discarded; branching over the surviving placements cannot recover them. For example, growing fragment A across an edge (u, v) can exclude a placement that would permit a different neighboring fragment B. Omitting (u, v) in a sweep run removes that adjacency from both growth and fragment-preservation tests, potentially changing where growth stops and which placements survive. The union of these runs therefore explores alternative fragment boundaries while retaining the uncut result. Each run remains greedy, and a single-edge sweep cannot explore every possible partition. Cut-specific seed orderings also contribute trajectory diversity, so the measured gain reflects the complete sweep schedule. Final bond events are always evaluated on the original, uncut endpoint matrices.

SLAP augmented with our single-edge sweep recovers 1,796/1,851 references (97.03%) using both directions and binary and weighted modes. Relative to this strongest evaluated prior-method baseline, one-order GRAFT has a higher verified recovery rate by 2.05 percentage points. The paired comparison contains 43 references verified only by one-order GRAFT and 5 verified only by SLAP. This compares recovery among alternatives under the stated configurations; it is not a comparison of single-answer accuracy or equal-size candidate sets.

On the same-Mac subset of 1,821 completed reactions, the one-seed GRAFT sweep-search stage uses a mean 1.14 CPU seconds for smaller-first search, 1.38 for larger-first, and 2.52 for the bidirectional union, compared

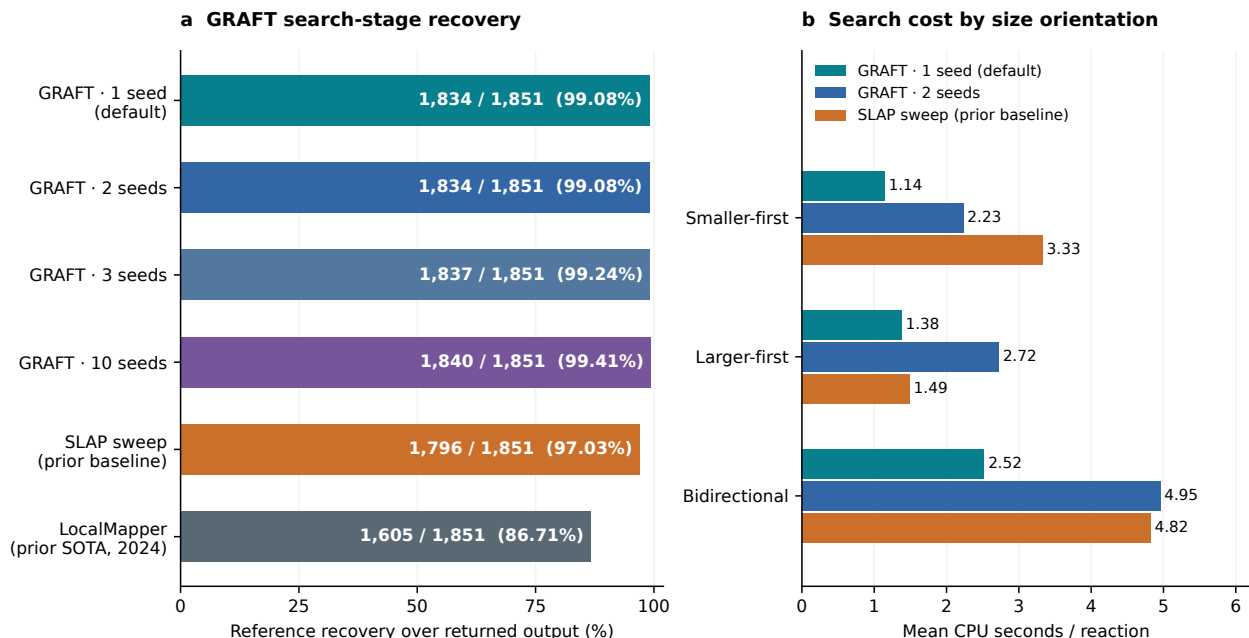


Figure 2 Reference recovery and search cost. GRAFT measurements cover sweep search. (a) Golden recovery for GRAFT and SLAP with our cut-sweep strategy, with LocalMapper as the prior accuracy-SOTA baseline (2024). LocalMapper returns one bijection; the other bars measure coverage among alternatives. All values use our evaluator, not literature-reported accuracies. Unswept controls are reported in Table 2. Every percentage uses all 1,851 records. (b) Mean CPU for smaller-first, larger-first, and bidirectional search on the same 1,821 completed reactions on the Mac. Reference verification is excluded; the procedures have different search objectives, output cardinalities, and I/O timing scopes.

with 3.33, 1.49, and 4.82 for the corresponding SLAP sweeps (Table 3 and figure 2). Two-seed GRAFT costs 2.23, 2.72, and 4.95 CPU seconds. Thus, orientation and seed count both affect the recovery–cost tradeoff. Its bidirectional median is 0.72 seconds versus SLAP’s 0.88, but its 95th percentile is higher: 12.91 versus 11.05 seconds. The one-seed bidirectional mean is 0.52 times the SLAP-sweep baseline (47.7% lower), whereas two-seed GRAFT costs 1.03 times that baseline. Lower means and medians do not imply uniformly lower runtime.

Golden runtimes measure search, excluding reference verification and separate bond-event decoding. GRAFT includes checkpoint I/O and symmetry finalization; SLAP includes graph construction, mapping, and export but excludes I/O. Bidirectional CPU sums both directions. The controlled timing cohort in Table 2 provides costs for every GRAFT seed setting and both no-sweep controls on the AMD EPYC 9J14 CPU. These values are separate from the Apple M4 Pro orientation measurements above. The table retains primary-campaign recovery counts; the controlled rerun’s own unresolved outcomes are reported in Table A3. In the primary evaluation, the uncut and swept bidirectional SLAP searches have one and three unresolved reactions, respectively; all GRAFT recovery verdicts are resolved. Archived default-call CPU times appear alongside recovery in the same table; detailed wall times and call counts remain in the repository.

3.3 Comparison with released mapping implementations

We also evaluate default released implementations of RXNMapper (11), LocalMapper (2), Chython (3), Indigo (6), and the Reaction Decoder Tool (RDT) (10), together with default binary SLAP. We select binary SLAP because it has higher reference recovery than the weighted default in our evaluation; both configurations remain in the archived results. Each receives all 1,851 label-free Golden records. Saved predictions are rescored with the same strict heavy-atom relation criterion, retaining unmatched atoms and checking endpoint chemistry. No reference-guided reranking, balancing, or prediction repair is applied.

Table 3 Recovery and runtime by endpoint-size orientation. All rows use the sweep. Recovery uses all 1,851 Golden records; mean, median, and 95th-percentile CPU seconds per reaction use the same 1,821 completed Mac reactions. SLAP is the prior-method baseline and unions binary and weighted modes within each direction. CPU/SLAP divides each mean by the SLAP mean for that same orientation; values below one indicate lower mean cost. Size counts explicit atoms; equal-size ties use stored order for smaller-first and its reverse for larger-first. Bidirectional cost sums both searches, not parallel wall latency. GRAFT includes checkpoint I/O and symmetry finalization; SLAP includes graph construction, mapping, and export but excludes I/O. Reference verification is excluded.

Method	Orientation	Recovered	Mean	Median	95th pct.	CPU/SLAP
GRAFT, 1 seed (default)	Smaller-first	1,803 (97.41%)	1.141	0.263	5.289	0.34
	Larger-first	1,779 (96.11%)	1.377	0.348	6.401	0.92
	Bidirectional	1,834 (99.08%)	2.518	0.719	12.913	0.52
GRAFT, 2 seeds	Smaller-first	1,815 (98.06%)	2.232	0.416	10.635	0.67
	Larger-first	1,805 (97.51%)	2.721	0.578	14.064	1.83
	Bidirectional	1,834 (99.08%)	4.953	1.250	26.350	1.03
SLAP sweep (prior baseline)	Smaller-first	1,775 (95.89%)	3.329	0.422	4.752	1.00
	Larger-first	1,770 (95.62%)	1.490	0.455	4.568	1.00
	Bidirectional	1,796 (97.03%)	4.819	0.881	11.048	1.00

Table 2 compares GRAFT with every benchmarked implementation using reference recovery over the evaluated output. GRAFT ranges from 1,489 recoveries without the sweep to 1,834 with a one-seed sweep and 1,840 with ten seeds. LocalMapper recovers the strict reference relation in 1,605 reactions, RXNMapper in 1,547, and Chython in 1,561. Default binary SLAP recovers 1,591 among its explicit candidates, compared with 1,796 for the broader bidirectional sweep evaluated above. These default calls use different search budgets from GRAFT; GRAFT’s family recovery is not a top-one accuracy measurement.

RDT recovers 1,030 references under this relation-based criterion, which differs from its chemical-change scoring. Indigo’s 692 recoveries require particular care: 503 outputs contain duplicate heavy-atom map labels, seven alter endpoint chemistry, and 133 calls fail. Chython changes endpoint chemistry in 29 outputs, and three RDT outputs fail validation. These results characterize the pinned implementations and input/output protocol rather than establish universal rankings of the methods. Exact versions and reproduction settings are recorded in the repository.

3.4 Minimum bond-event comparison on 140 coordinate reactions

We compare the fewest bond events found for each reaction and the different bond-change patterns that achieve that count. An event count tells us how many atom pairs change; a pattern specifies which pairs gain or lose bond order. We score all mappings on the same original WBO matrices using the thresholds in Equation (5). GRAFT uses the default pipeline: one-seed sweep, with branch cap 100 by default and 2,000 as a larger-budget comparison. We compare it with released SLAP (native SLAP) and SLAP augmented with our cut sweep (SLAP sweep). These reactions have no verified atom mappings, so this measures agreement and differences in bond events, not mapping accuracy. Throughout this section, a minimum is the lowest count found by a method, not a proven global optimum.

The coordinate comparison uses cap 2,000, which returns complete mappings for all 140 reactions. The archived paired search control at cap 100 returns complete mappings for 138 reactions; cases 123 and 125 return no complete mapping. These are search-budget failures. Their cap-2,000 outputs contain three- and four-event solutions, respectively. The library default remains cap 100; the detailed coordinate results below explicitly use the larger cap. Golden includes a separate cap ablation in Appendix A.6.

At cap 2,000, GRAFT reaches the same minimum count as SLAP sweep in 136 reactions and one fewer event in the remaining four. Against native SLAP, it reaches the same count in 125 reactions and fewer events in 15 (Table 4 and figure 3). Thus, GRAFT matches or lowers the event count in every reaction under both comparisons. A lower count alone does not establish a more plausible chemical transformation.

Equal counts can still describe different bond changes. Among the 136 reactions where GRAFT and SLAP sweep have equal minima, SLAP sweep returns 164 minimum-event patterns; GRAFT recovers 158. The six missing patterns occur in four reactions: one each in cases 6, 11 and 101, and three in case 64. For the 125 reactions where GRAFT and native SLAP have equal minima, GRAFT recovers 138 of 140 native-SLAP patterns. Across all 140 reactions, the number of patterns at each method’s own minima is 166 for GRAFT, 168 for SLAP sweep, and 160 for native SLAP. These totals count patterns, not reactions.

Table 4 Minimum bond-event counts and patterns on 140 reactions at cap 2,000. The first three numerical columns count reactions where GRAFT finds fewer, equal, or more events than the comparator. The final column counts comparator patterns recovered by GRAFT, restricted to reactions with equal minima. For example, 158/164 refers to patterns across the 136 reactions in the preceding column. Settings are given in the text.

Compared with	GRAFT fewer	Equal	GRAFT more	Patterns at equal minima
SLAP sweep	4	136	0	158/164
Native SLAP	15	125	0	138/140

We also check whether GRAFT retains SLAP’s minimum-event patterns when GRAFT finds a lower count of its own. For each reaction, we decode GRAFT candidates up to one event above the higher of the two SLAP minima. Within this range, GRAFT recovers 162 of 168 SLAP-sweep patterns and 151 of 160 native-SLAP patterns. Its output contains 300 distinct patterns: 166 at GRAFT’s minima and 134 with higher counts. Decoding is complete within this range for the saved families; the fragment search itself is not exhaustive.

Table 5 CPU time per reaction for the cap-2,000, 140-case comparison. All stages were measured on the Mac. The GRAFT row measures sweep search; separate event decoding is not timed here. SLAP refinement covers its returned witnesses. These are stage costs, not directly comparable end-to-end runtimes. GRAFT search includes checkpoint I/O. SLAP mapping excludes input preparation and export, while its refinement includes I/O. Final shared event rescoring and failed or replaced attempts are excluded.

Recorded stage	Mean CPU s	Median CPU s	95th pct. CPU s
GRAFT sweep search	1.047	0.314	3.481
SLAP sweep mapping	1.337	0.546	5.919
SLAP H refinement / initial scoring	0.053	0.011	0.207

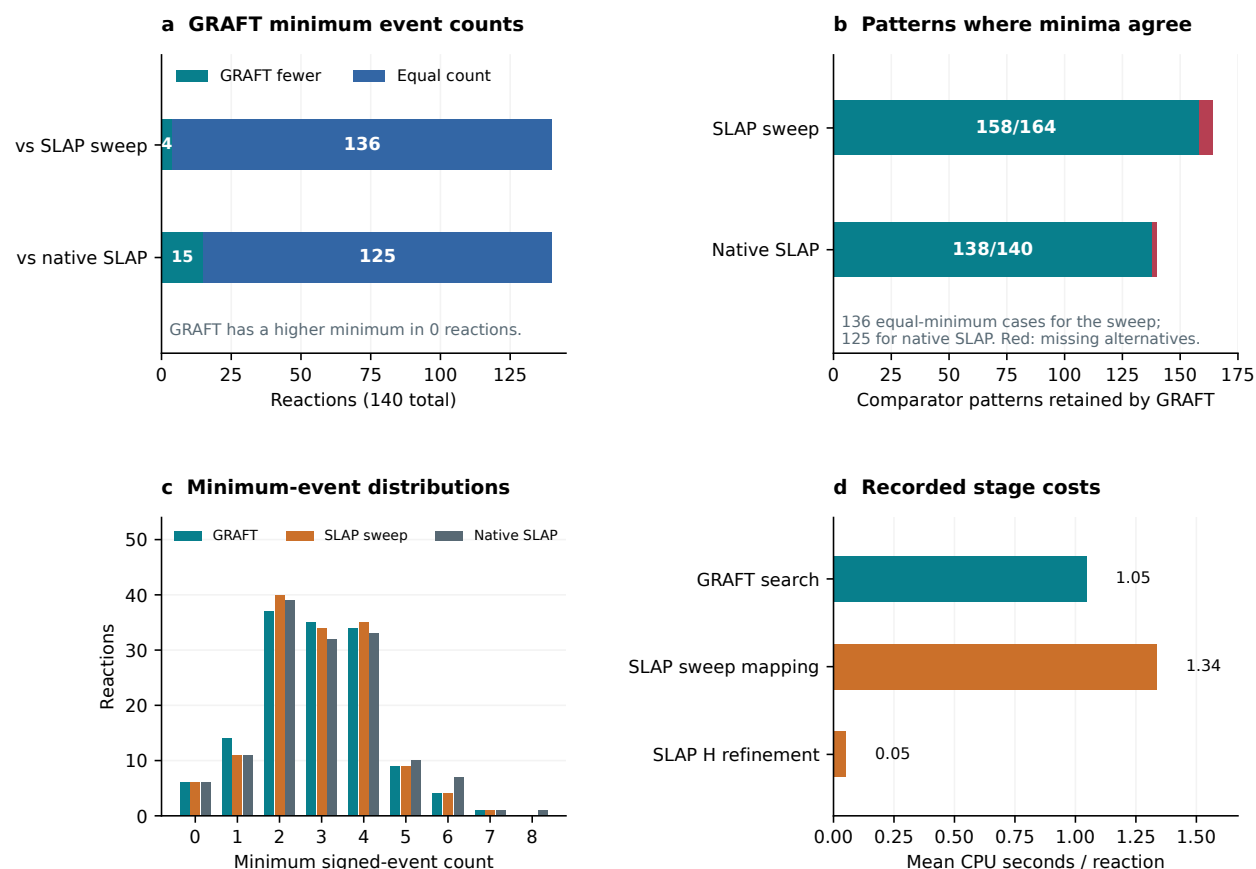


Figure 3 Bond-event counts, alternative patterns, and runtime on 140 coordinate reactions. GRAFT uses one seed and the sweep at cap 2,000. (a) Reactions where GRAFT finds fewer or equal events than each SLAP baseline; none has a higher GRAFT minimum. (b) Comparator minimum-event patterns recovered by GRAFT, counting only reactions where both methods reach the same minimum. The six missing SLAP-sweep patterns span four reactions. (c) Distribution of the minimum event count across reactions for each method. (d) Mean CPU seconds per reaction for the recorded stages. The timing panel reports search and SLAP refinement; it excludes GRAFT event decoding. See Table 5 for timing scope. The mappings are unverified, so this figure compares bond events rather than mapping accuracy.

Sweep search takes a mean 1.05 CPU seconds per reaction. SLAP sweep takes 1.34 CPU seconds for mapping and 0.053 for hydrogen refinement and initial scoring. GRAFT’s separate decoder reports all retained event classes in the specified windows, whereas SLAP refinement covers its returned witnesses; these timings therefore do not imply an end-to-end speed ranking. We did not separately time native SLAP or minimum-only GRAFT decoding.

A fresh run of the final implementation completed search and event decoding for 140/140 coordinate reactions. On this full set, the full pipeline takes a mean 6.07 CPU seconds per reaction (median 0.81; 95th percentile 25.47). Decoding accounts for a mean 4.65 CPU seconds (median 0.29). These measurements include search, final-family construction, decoding, and candidate output; they are separate from the search-only comparison above (Table A4).

3.5 Final representation and complete window decoding

Unordered grouping collects histories with the same matched fragment pairs while retaining their constrained mapping families. Every saved baseline family is decoded or certified within its specified event window. The resulting 300 reaction-specific classes each have a mapping witness. This certifies the saved output within the window, not exhaustive structural search. Counts and per-family certificates are provided with the code.

4 Discussion

GRAFT separates structural search, symmetry representation, and interpretation of bond changes. Continuous growth accumulates context while preserving multiple placements; the cut sweep varies the constraints that determine fragment boundaries. Final grouping removes irrelevant growth order, while the flat family representation retains constraints and correlated permutations that can affect the output. Symbolic projection then exposes distinct bond-event hypotheses without expanding every mapping.

4.1 What a single correspondence leaves out

One correspondence is sufficient when atom identities are established and the downstream quantity is invariant over the remaining alternatives. Otherwise, selecting one witness discards two kinds of information: different assignments can produce different bond-event patterns, and a chosen pattern can still admit joint atom shuffles. Formally, if $\mathcal{F}$ is the retained mapping set and $e(m)$ its event-class projection, a witness m_0 supplies $e(m_0)$ but cannot reconstruct $\{e(m) : m \in \mathcal{F}\}$ whenever that set has more than one member. Event counts alone are also insufficient: distinct signed patterns can have equal counts. GRAFT therefore returns an event-class representative together with access to its supporting families and conditional shuffle queries.

Golden case 9 illustrates why retaining alternatives matters (Figure 4). GRAFT recovers the annotated heavy-atom correspondence: the alcohol oxygen O_{15} enters the product ring, while the ketone oxygen O_3 leaves in water (A). A second recovered mapping exchanges these oxygen origins (B), and a third additionally exchanges the correspondence of C_{13} and C_{14} (C). The three displayed witnesses have five, six, and seven heavy-atom changes, respectively; including H gives nine, eight, and nine signed events. Each witness attains the minimum number of H-involving changes for its fixed heavy-atom correspondence, as verified from the attached-H counts. Thus a lower total event count can favor a nonreference mapping, while equal counts (A and C) do not establish equal correspondences. Golden annotates heavy atoms, not H identities or a validated pathway. These are selected recovered witnesses, not an exhaustive decoding of this reaction.

The output distinction is specific to the interfaces evaluated here. RXNMapper, LocalMapper, Indigo, Chython, and RDT returned at most one explicit candidate record per reaction in the archived runs; SLAP exposed multiple records and compressed labeled-graph information, while GRAFT exposes conditional mapping families. This does not establish that only SLAP and GRAFT can return alternatives. For example, Ali et al. enumerate optimal mappings with an Ising-computing approach and reduce them by molecular symmetry (1). Nor does high reference recovery establish near-complete coverage of all mappings: GRAFT’s structural search is bounded, and complete decoding certifies only the saved families within the requested event window.

4.2 Implications for pathway and transition-state analysis

Different bond-event patterns can suggest different reaction centers and alternative transition-state starting guesses. They do not identify pathways uniquely: a single net atom mapping can be compatible with several intermediate sequences, and an alternative mapping can be chemically implausible. Energies, reaction conditions, stereochemistry, and isotope tracing can distinguish possibilities that endpoint connectivity alone cannot. Prior work on reaction-pathway search also identifies atomic-index assignment as a practical obstacle and addresses the factorial number of index assignments without enumerating them (4).

For a transition-state pipeline, the relevant choices are the event-class candidate and the allowed atom assignments around its reaction core. Define the core as the atoms incident to the candidate’s signed bond events. Alternative assignments of those atoms can be queried jointly against the saved family constraints, retaining only assignments that extend to a complete mapping and satisfy the chosen event-class restriction. Required correlated movements of other atom assignments must be included in that completion. The completed mapping determines the product atom ordering supplied to the interpolation step.

Different completed mappings can then give different interpolations between the same endpoint structures. For illustration, in a common coordinate frame, a Cartesian interpolation for mapping m is

$$\mathbf{x}_i^{(m)}(t) = (1 - t)\mathbf{x}_i^R + t\mathbf{x}_{m(i)}^P, \quad 0 \leq t \leq 1. \quad (6)$$

Golden case 9: the reference is one of several recovered mappings

GRAFT search · 1 seed (default) · cut sweep · cap 2,000 · iso tolerance 1 · event threshold 0.5

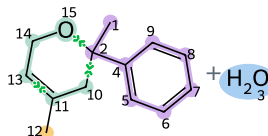
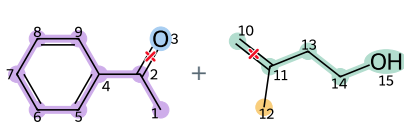
● ● ● Saved fragment groups

—×— Break / weaken

—>— Form / strengthen

A Ground truth (Golden reference) — recovered by GRAFT

5 heavy-atom changes

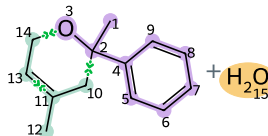
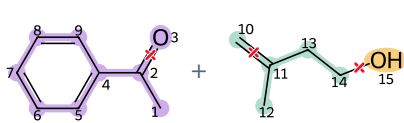


Reference: alcohol O₁₅ enters the ring; ketone O₃ leaves in water.

9 total events = 5 heavy + 4 involving H

B GRAFT alternative — different oxygen retained

6 heavy-atom changes

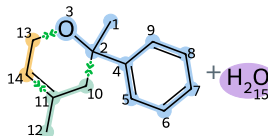
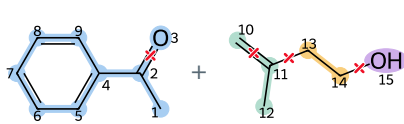


Alternative: ketone O₃ enters the ring; alcohol O₁₅ leaves in water.

8 total events = 6 heavy + 2 involving H

C GRAFT alternative — oxygen and carbon correspondence change

7 heavy-atom changes



Alternative: O₃ enters the ring, with a different assignment of C₁₃ and C₁₄.

9 total events = 7 heavy + 2 involving H

Retaining alternatives matters: a lower total event count (B: 8) need not select the annotated mapping (A: 9).

Ground truth refers to Golden's heavy-atom annotation. H identities are unannotated; totals use each GRAFT witness.

Numbers are original Golden labels. H atoms are implicit. Selected witnesses do not imply exhaustive or minimum-event decoding.

Figure 4 Golden reference and recovered GRAFT alternatives for case 9. A matches all 15 annotated heavy-atom pairs exactly. B exchanges the oxygen origins; C also exchanges C₁₃/C₁₄. The GRAFT sweep-search stage uses one seed (default), a 2,000-branch cap, and $\tau_{\text{iso}} = 1$; reference labels are removed before search. Pastel colors identify saved fragment groups, consistently across each row's endpoints; partitions can differ between rows. Solid black lines retain the endpoint bonds. Red × marks indicate cleavage or weakening, and paired green arrows pointing inward indicate formation or strengthening. The marks sit directly on the affected bonds; fragment colors remain visible. Events use formal bond orders at threshold 0.5. Numbers are the original Golden atom labels. H atoms are implicit; the figure separates heavy-atom changes from totals including H. "Ground truth" denotes the dataset's heavy-atom annotation, not a verified pathway. Only three selected witnesses are shown.

Changing $m(i)$ changes the product coordinate assigned to reactant atom i , and hence generally changes its interpolated trajectory. Retaining multiple event classes and their allowed core assignments therefore lets a downstream pipeline construct and assess several initial paths instead of committing to one correspondence before interpolation. Equivalent relabelings applied consistently to an entire calculation do not create new physical pathways; the issue is the relative correspondence between fixed endpoints. These examples motivate retaining alternatives and accessible symmetry information; transition-state search performance is not evaluated here.

4.3 Coverage and remaining search limitations

The Golden results support broad reference coverage by the fragment-search stage, and the coordinate comparison shows that GRAFT reaches equal or lower event counts than the returned SLAP-sweep candidates while retaining most patterns at shared minima. These are complementary measurements. The coordinate collection has no mapping annotations, and event minimization is not an energy model or a proof of mecha-

nistic plausibility. The coordinate collection was inspected during development, so independent validation is needed to establish generalization.

An audit of all eleven ten-order Golden misses separates unmatched-atom decisions from fragment-placement failures (Table A2). In four reactions, every retained family maps exactly one more heavy-atom pair than the reference. Such a family cannot reproduce the reference’s unmatched-atom pattern through symmetry shuffling: permutations preserve the number of assigned pairs. The remaining seven lack the required correspondence within their saved families. Re-ranking or expanding those families into explicit bijections cannot supply a missing structural alternative.

Two traced examples show how greedy extension excludes a representable reference. One reference exchanges adjacent labels in an otherwise unchanged pyridine ring; another exchanges the carbonyl-oxygen attachments of oxalic acid across its retained carbons. Ordinary growth preserves bonds that these correspondences require separating. Uncapped trajectories still miss both references, while reference-constrained fragment choices establish their representability. In the oxalic-acid example, a reference-selected three-edge relaxation also exposes the reference to ordinary matching; the single-edge sweep does not contain that cut set. These are feasibility diagnostics, not additional benchmark recoveries. For the other five correspondence misses, the audit does not isolate cap truncation from unsampled growth choices. They motivate selective alternatives to successful growth extensions and explicit choices about leaving atoms unmatched, rather than treating larger caps or more symmetry expansion as sufficient remedies.

Only one of the eleven misses is recovered by any tested comparator configuration, including the SLAP sweep. The shared failures warrant inspection of the supplied correspondences as well as search improvements; they do not establish annotation errors. All eleven remain in the reported denominator. A complete event-window certificate likewise applies only to the saved families, whose matching-preserving automorphisms can still change finer-threshold events.

Code and data

Code and reproduction instructions are available at https://github.com/yunhzou/coordinate_alignment, branch `main`. The repository guide `manuscript/REPRODUCIBILITY.md` links full benchmark configurations, software versions, per-case results, and figure-generation scripts. Golden is distributed by its original authors (8).

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A Evaluation details

A.1 Reproducibility and comparison scope

The repository’s `manuscript/REPRODUCIBILITY.md` links the benchmark scripts, pinned software and input versions, full configurations, and per-case records. [Table 1](#) gives the settings needed to interpret the comparisons. The notebook `docs/AAM_SIMPLE.ipynb` demonstrates matching, unique event candidates, and conditional symmetry queries through the Python API.

For the coordinate comparison, GRAFT uses supplied WBO matrices and SLAP uses the original XYZ-derived binary graphs. SLAP’s returned heavy-atom assignments receive hydrogen refinement under its original bond-change objective, then are rescored on the WBO matrices. Thus, the comparison uses a common event score but different input representations. It covers the returned SLAP witnesses, not an exhaustive expansion of its hydrogen label families. The WBO-generation method is not recorded in the available metadata, so regeneration of these matrices from molecular structures is not established.

The paired search timings use the Apple M4 Pro CPU and one numerical-library thread. Averages use 1,821 common completed reactions. Reference verification and interrupted work are excluded. GRAFT includes checkpoint I/O; SLAP excludes I/O. Higher-seed runs measured on different CPUs are excluded from this paired comparison. Stage-specific timing scopes are given with the corresponding tables.

A.2 Default comparator protocol

Released comparators receive the same label-free Golden inputs, without prediction repair or reference-guided ranking. Invalid outputs and failures remain in the denominator. SLAP is evaluated over all returned candidates, with binary and weighted modes reported separately. The repository records the released models, versions, and adapter settings.

A.3 Decoding scope

For each coordinate reaction, the decoding limit is one event above the larger of the native-SLAP and SLAP-sweep minima. All 120,052 retained families are exhausted within these limits, without a class-count cap. Higher-event patterns are outside this completeness claim. An event class records the signed atom-pair changes, modulo source permutations preserving every thresholded bond-change test against the opposite endpoint. Each class has a concrete mapping witness.

Table A1 Complete final decoding on all 140 reactions. Every retained family is certified within the listed per-case event limit.

Maximum events	1	2	3	4	5	6	7	8	9
Reactions (all complete)	6	11	39	32	33	10	7	1	1

A.4 Why flattening preserves the saved mappings

Flattening retains each terminal representative, its required fragment constraints, and its allowed action sequence after applying prefix locks. Disjoint actions commute. Adjacent nested groups satisfy $HG = GH = G$ when $H \leq G$, with containment checked by exact generator membership. Other overlapping actions retain their order. Identical records are shared; branches with the same unordered fragment pairs retain the union of their family records. The admissible mapping union is therefore unchanged. A shared fragment partition or automorphism group alone does not establish family equality.

The decoder certifies a family through event invariance, a valid lower bound excluding the requested window, or unsatisfiability after excluding all discovered event classes. An exhausted budget remains incomplete. This separates lossless representation, complete decoding within a window, and exhaustive structural search.

A.5 Golden nonrecoveries

Table A2 The eleven Golden nonrecoveries. Case indices are zero-based. Retained pair counts cover every terminal in both directions and every saved cut at ten seed orderings and cap 100. The four cardinality mismatches concern exact reference relations, including unmatched atoms.

Case	Reference pairs	Retained pairs	Observed mismatch
7	29	30	One extra matched pair
19	23	23	Atom pairing excluded by orbit bounds
590	31	31	Atom pairing excluded by orbit bounds
871	20	20	Atom pairing excluded by orbit bounds
986	16	16	Atom pairing excluded by orbit bounds
1228	12	12	Atom pairing excluded by orbit bounds
1285	14	14	Joint C/O correspondence excluded
1358	36	37	One extra matched pair
1377	22	23	One extra matched pair
1475	18	19	One extra matched pair
1553	20	20	Atom pairing excluded by orbit bounds

Independent reconstruction agrees with the supplied references in all eleven cases. Four require a different number of matched pairs; seven require correspondences absent from the retained families. All have at least one capped direction, but this does not identify the cap as the cause. The uncapped traces discussed in the main text demonstrate greedy-growth exclusions in two cases. Reference-directed feasibility checks are diagnostic and do not contribute to benchmark recovery. Per-case certificates and traces are linked in the repository.

A.6 Branch-cap and CPU-time ablation on Golden

We reran the twelve configurations in [Table A3](#) on all 1,851 Golden reactions using one CPU thread on an AMD EPYC 9J14 processor. Both search directions are unioned. SLAP unions its binary and weighted modes. Search and reference verification have separate fixed resource budgets. The recorded cap-100 recovery counts reproduce the main evaluation. At cap 2,000, unresolved counts increase with seed count. Among reactions resolved at both caps, no swept configuration gains or loses a reference recovery.

The ten-seed cap-2,000 run leaves five references recovered at cap 100 unresolved, accounting for its lower confirmed count. The bounded SLAP sweep confirms 1,794 references here, compared with 1,796 in the main evaluation: two previously recovered cases are unresolved under this study’s limits. These outcomes are reported separately from verified exclusions. The paired timing cohort excludes 448 reactions because at least one configuration does not finish; the all-attempt cost column exposes the additional computational cost on those difficult reactions. Source pins, individual outcomes, directional costs, and reproduction scripts are provided with the code.

Table A3 Golden branch-cap ablation with paired SLAP baselines. Recovery counts confirmed reference inclusion over all 1,851 records; unknown cases remain in the denominator. CPU values are seconds per reaction. Paired CPU uses the same 1,403 reactions whose searches finish under every configuration. Spent CPU averages recorded search cost over all 1,851 selected attempts, including interrupted calls; it is not completed-search latency. Both timings exclude reference verification and initial imports. Superseded attempts are excluded. Cap 100 with a one-seed sweep is the default.

Method	Cap	Seeds	Recovered (%)	Unknown	Paired CPU	Spent CPU
GRAFT uncut	100	1	1,489 (80.44)	0	0.040	0.085
GRAFT sweep (default)	100	1	1,834 (99.08)	0	0.963	3.873
GRAFT sweep	100	2	1,834 (99.08)	0	1.619	7.260
GRAFT sweep	100	3	1,837 (99.24)	0	2.265	10.556
GRAFT sweep	100	10	1,840 (99.41)	0	6.993	33.211
GRAFT uncut	2,000	1	1,508 (81.47)	5	0.086	0.517
GRAFT sweep	2,000	1	1,834 (99.08)	2	4.317	30.055
GRAFT sweep	2,000	2	1,834 (99.08)	4	8.283	48.549
GRAFT sweep	2,000	3	1,837 (99.24)	6	12.201	61.890
GRAFT sweep	2,000	10	1,835 (99.14)	12	43.843	113.777
SLAP uncut	–	–	1,661 (89.74)	1	0.105	0.659
SLAP sweep	–	–	1,794 (96.92)	5	2.793	9.895

A.7 Fresh end-to-end timing on coordinate reactions

We measured one-seed, cap-2,000 cut-sweep search followed by the public event decoder on the Apple M4 Pro CPU. Four reactions ran concurrently, with one numerical thread per reaction. CPU times are per reaction, not divided by the worker count. The event thresholds and windows match the bond-event comparison, with no class-count cap. All 140 completed reactions reproduce the archived event-class sets and family counts. Median end-to-end wall time is 0.96 seconds. Timing includes input loading, checkpoint I/O, final-family construction, decoding, candidate serialization, and progress recording; initial imports and reference-set verification are excluded. All 140 reactions finish within the five-minute watchdog. Per-case timings, source hashes, and reproduction commands are provided with the code.

Table A4 GRAFT search plus bond-event decoding. CPU seconds per completed reaction in the fresh 140-reaction pass. The final row measures the whole pipeline; medians and percentiles of individual stages are not additive. Completion coverage and timing scope are stated above.

Stage	Mean CPU s	Median CPU s	95th pct. CPU s
GRAFT sweep search	1.257	0.300	3.468
Final-family catalogue	0.156	0.019	0.595
Bond-event decoding	4.647	0.288	24.361
Candidate serialization	0.005	0.001	0.015
End to end	6.073	0.812	25.470